

Ward Boundary Feasibility

1. Source Check

Official AMC GIS portal: exists, but doesn't offer an easy direct-download boundary file for programmatic use.

Working secondary source found: DataMeet India's open community GIS repository — github.com/datameet/Municipal_Spatial_Data/Ahmedabad/Wards.geojson. This is a well-known, citable open-data source (CC BY 4.0 license) used by Indian civic-tech and research projects.

2. Core Test: "Can we load actual ward polygons in Python?" — YES

- File downloaded and loaded successfully with `geopandas.read_file()`
- **48 ward polygons confirmed** — exactly matches AMC's official ward count
- CRS: EPSG:4326 (standard lat/lon, no reprojection needed)
- Bounding box matches Ahmedabad's real location (72.45–72.70°E, 22.91–23.14°N)
- Each ward has a Name field (e.g., "48 RAMOL HATHIJAN") — cleaned into a stable Ward_ID (1–48) and a readable Ward_Name column for joining with thermal/demographic data later

Real plot generated (`ahmedabad_wards_plot.png`) — visually confirms all 48 ward shapes render correctly and cover the expected city extent.

3. Known data-quality note

A few faint diagonal line artifacts are visible across some central wards in the plot — a minor boundary-drawing imperfection in the community-sourced file, not uncommon in open GIS data. Doesn't block spatial joins (each ward polygon is still valid and distinct), but worth a quick visual QA pass before the final map goes into the dashboard, and worth noting honestly in the Decision Log if asked about data quality.

4. Demographic data availability

Already confirmed via Khushi's Day 1 research note: AMC's own website (ahmedabadcity.gov.in/SP/Demographics) provides a ward-wise Census 2011 download (population, male/female, SC/ST, literacy, 0–6 age group). Elderly (60+) breakdown and outdoor-worker data still need verification once that file is actually opened — flagged as 🟡 in Khushi's note.

5. Files produced

- `ahmedabad_wards_clean.geojson` — cleaned ward boundaries with stable `Ward_ID` + `Ward_Name`, ready to join with thermal/demographic data
- `ahmedabad_wards_plot.png` — proof-of-work visual
- Source script for reproducibility: `load_ward_boundaries.py`

Handoff

- `ahmedabad_wards_clean.geojson` → goes to **Vrinda** for the dashboard map
- `Ward_ID` scheme → this is the join key Dev/Shweta's thermal dataset and the demographic dataset will both use later

Answer to Day 1's core question: ward-level spatial data feasibility is CONFIRMED, not just promising.